

Transition: DS + Algorithms

Linear
recurrences



Recursive algorithms

A function which calls itself

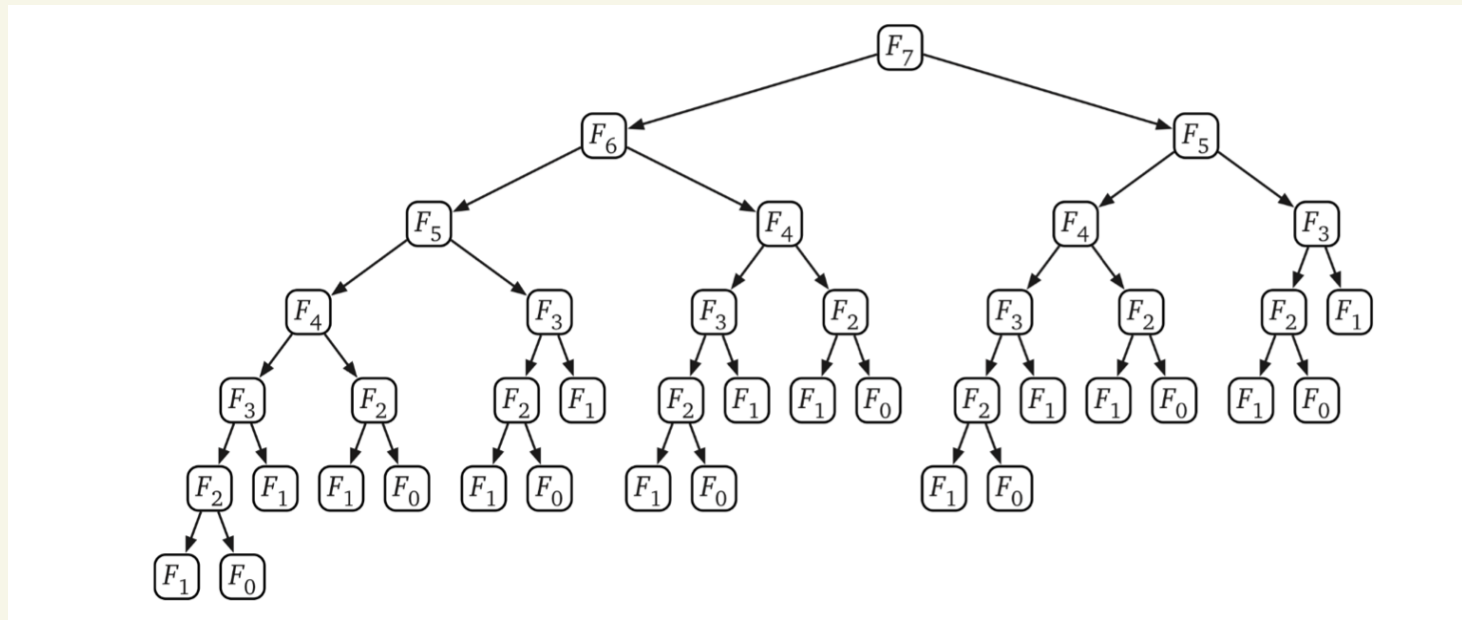
↳ but, with smaller input!

Example: $f_n = f_{n-1} + f_{n-2}$

↳ FIB(n):
if $n < 2$:
 return n
else
 return $FIB(n-1) + FIB(n-2)$

Unpacking the recursion

Call $\text{Fib}(7)$:



Note:

Not all happening at once!
check code & think about how
programs work etc.

Why?

Well, for more complex algorithms/structures
can use a recursive definition to
build the algorithms

↳ then use induction to prove
correctness! (see last reading)

Also $\rightarrow$ runtime often becomes a
recurrence.

So, need to do big-O , but
of a recurrence!

Recall: $n! =$

How to compute?

FACTORIAL(n):

if $n = 0$
return 1

else

$r :=$

return

Correctness:
Proof.

induction on n :

Base case:

Ind Hyp:

Ind Step:

Runtime:

Let $T(k)$ = runtime of function on input k .

Then:

$$T(n) =$$

"Unrolling":

When unrolling fails (& it often does):

FIB(n):

if $n < 2$:

return n

else

return $FIB(n-1) + FIB(n-2)$

Recurrence:

Defn A linear homogeneous recurrence has the form

$$a_n = C_1 a_{n-1} + C_2 a_{n-2} + \dots + C_d a_{n-d}$$

where $C_1, \dots, C_d$ are constant and $C_d \neq 0$

The order of the recurrence is d .

Examples:

$$a_n = 3a_{n-1} + 2a_{n-2}$$

$$T(n) = 4T(n-4) + 11$$

$$F_n = F_{n-1} + F_{n-2}$$

Examples: Yes or no?

$$P(n) = 1.06 P(n-1)$$

$$H(n) = 2H(n-1) + 1$$

$$R(n) = R(n-1) + (R(n-2))^2$$

$$a_n = \sqrt{3} a_{n-5}$$

$$f_n = n \cdot f_{n-1}$$

$$M(n) = 2M\left(\frac{n}{3}\right) + n$$

How to solve:

Generally, these types have solutions which are exponential.

So $\rightarrow$ look for equations of the

$$\text{form } a_n = r^n$$

where r is some constant

If a solution of this form exists,

$$\text{have: } a_n = C_1 a_{n-1} + C_2 a_{n-2} + \dots + C_d a_{n-d}$$

$$\Downarrow$$
$$r^n = C_1 r^{n-1} + C_2 r^{n-2} + \dots + C_d r^{n-d}$$

divide common term:

So: The sequence $a_n = r^n$ is a solution



r is a solution of the
characteristic eqn:

$$r^d - c_1 r^{d-1} - \dots - c_d = 0$$

How to find r ?

Ex: $P(n) = 1.06 \cdot P(n-1)$

Ex $a_n = a_{n-1} + 2a_{n-2}$

$$\text{Ex: } F_n = F_{n-1} + F_{n-2}$$

Ex: $B(n) = 2B(n-1) + B(n-2)$

... This is a bit different!

Finding general solutions

- If r is a non-repeated root of the char. eqn, then r^n is a solution to the recurrence.
- If r is a repeated root with multiplicity k then $r, n \cdot r^n, \dots, n^{k-1} r^n$ are all solutions
- Any linear combination will satisfy the recurrence, but need to use initial values to find constants

Huh?

Examples

- $P(n) = 1.06 P(n-1)$

$$P(0) = 10,000$$

$$x - 1.06 = 0 \Rightarrow \text{root is } x = 1.06$$

so $P(n) = C(1.06)^n$

what is C ?

$$\bullet A(n) = A(n-1) + 2A(n-2)$$

$$\bullet A(0) =$$

$$A(1) =$$

$$\Rightarrow x^2 - x - 2 = 0 \text{ had roots}$$

$$x = 2 \text{ and } x = -1$$

$$\text{So: } A(n) = C_1 \cdot 2^n + C_2 \cdot (-1)^n$$

- Suppose a characteristic equation for $C(n)$ simplifies to:

$$(x-2)^3 (x-5)^2 (x+9) = 0$$

Then roots are:

$$\text{and } C(n) \geq$$

So:

We can find runtimes, as long as
the recurrence is linear &
homogeneous!

Q: What if it's not??

Two others we'll study here,
since they are useful in computing.

- linear inhomogeneous

- divide & conquer

Linear inhomogeneous recurrences

These are just linear ones, with an added function $g(n)$ at end:

$$f(n) = c_1 f(n-1) + \dots + c_d f(n-d) + g(n)$$

Examples:

$$F(n) = F(n-1) + F(n-2) + 1$$

$$A(n) = 4A(n-1) - 2^n + n^2$$

Next time: how to solve?